

Physicochemical, Antioxidant, and Sensory Profiling of Commercial Immunonutritional Milk Supplements

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Please Ponder Upon...

What if the immunonutrition supplement with the strongest antioxidant potential is not the one patients are willing or able to drink?

01 WHY IMMUNONUTRITION MATTERS?

- Immunonutrition (IN) is a specialized nutritional approach designed to replenish essential nutrients and strengthen immune defense¹.
- Despite its therapeutic value, patient adherence is frequently compromised by poor palatability, making sensory acceptability a primary barrier to sustained use².
- Individual IN components exhibit antioxidant properties; the overall antioxidant capacity of IN formulas within a milk system is underexplored³.
- To evaluate the physicochemical properties, antioxidant profiles, and consumer sensory acceptability of commercial IN milks.

02 METHODOLOGY

5 coded commercial IN products (Products A, B, Neomune Vanilla [NV], D, and Neomune Melon [NM]).

Laboratory profiling

- Universiti Tunku Abdul Rahman (UTAR).
- October 2025–February 2026.
- Physicochemical:** Viscosity & Particle size.
- Antioxidant:** TPC, TFC, DPPH, FRAP & TAC.
- Data analysis was performed by ANOVA at $p \leq 0.05$ using SAS software.

Sensory profiling

- Cross-sectional study.
- Hospital Raja Permaisuri Bainun.
- May–June 2026.
- Convenience sampling:** 30 stable cancer patients & 30 healthy adults.
- Eligibility:** Adults aged ≥ 18 years who were able to taste and swallow IN, willing to provide informed consent, had no known allergies to formula ingredients & were stable cancer patients who were ≥ 3 months post-chemotherapy or radiotherapy.
- Appearance, aroma, taste, texture, aftertaste, and overall acceptability using a 5-point hedonic scale⁴.
- Data analysis was performed using SPSS software.

03 ANTIOXIDANT PROFILE

Functional assays identify different leaders

Figure 1: Total antioxidant capacity (mg AAE/g)

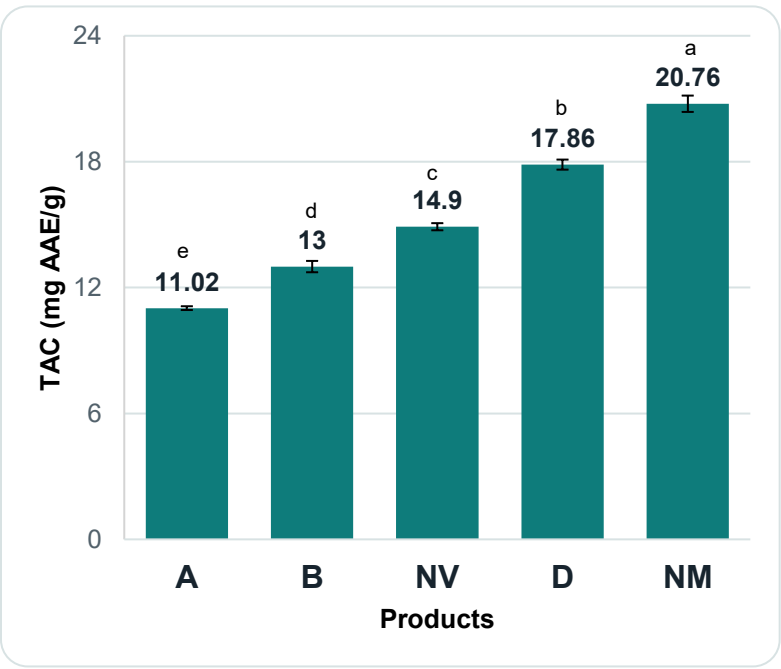
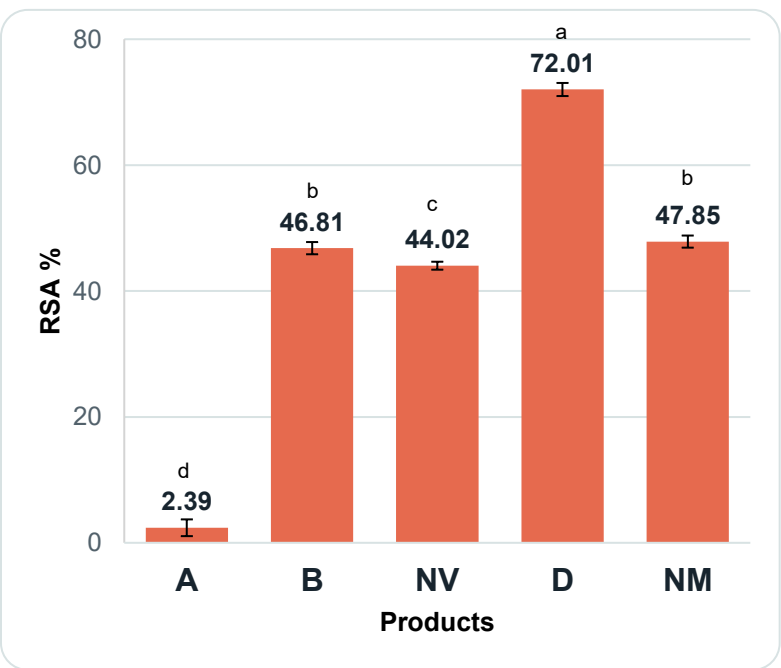


Figure 2: DPPH radical scavenging activity (RSA %)



Different assays, different leaders

- Product NM shows the highest TPC & TFC.
- Product D exhibits the highest DPPH & FRAP values, indicating the highest free radical scavenging and reducing power.
- Product NM exhibits moderate-to-strong antioxidant activity, with NM ranking the highest in total antioxidant activity.

Viscosity–antioxidant relationship

TPC: $r = -0.86$, $R^2 = 0.75$, $p < .0001$
TFC: $r = -0.88$, $R^2 = 0.76$, $p < .0001$

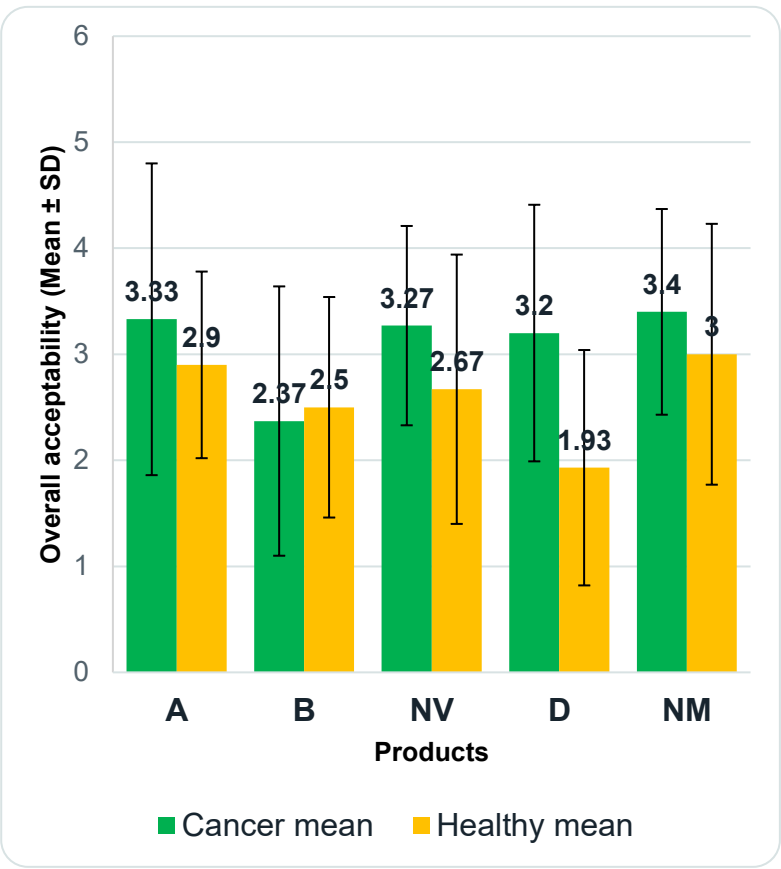
Higher viscosity was associated with lower total antioxidant content, TPC, and TFC content.

04 SENSORY ACCEPTABILITY

Table 1: Participants' Characteristics

	Healthy	Cancer
Age (years)	28.23 ± 12.08	66.97 ± 10.35
Gender	Female: 83.3% Male: 16.7%	Female: 50.0% Male: 50.0%
Ethnicity	Chinese: 63.3% Malay: 20.0% Indian: 16.7%	Chinese: 53.3% Indian: 46.7%

Figure 3: Product Ranking Based on Overall Acceptability (mean ± SD; n = 30/group)



Product NM ranked #1 in both populations

Cancer: 3.40 ± 0.97
Healthy: 3.00 ± 1.23

05 CLINICAL IMPLICATION

Effective clinical malnutrition management requires therapeutic formulations that align closely with patient palatability.

Balancing sensory appeal with robust physicochemical stability and antioxidant capacity is crucial for optimal immunonutrition.

These findings offer critical, evidence-based guidance for clinicians selecting appropriate oral supplements to maximize patient adherence and recovery.

06 CONFLICT OF INTEREST

None declared.

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